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Tuned mass absorber on a flexible structure

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ABSTRACT

The classic design of a tuned mass absorber is based on a simple two-mass analogy in which the tuned mass is connected to the structural mass with a spring and a viscous damper. In a flexible multi-degree-of-freedom structure the tuned mass absorber is typically introduced to provide damping of a specific mode. The motion of the point of attachment of the tuned mass absorber to the structure has not only a contribution from the targeted mode, but also a background contribution from other non-resonant modes. Similarly, the force provided by the tuned mass absorber is distributed between the targeted mode and the background modes. It is demonstrated how this effect can be included via a non-dimensional dynamic background flexibility coefficient, extracted from a classic modal analysis for the particular frequency of the selected mode. An explicit calibration procedure is developed starting with the desired maximum amplification, from which the device damper, mass and stiffness are determined, accounting for the background flexibility. Examples demonstrate the influence of the flexibility effect and the efficiency of the proposed procedure.

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1. Introduction

Lightly damped structures may develop large amplitude vibrations for loads acting near resonance frequencies. These vibrations can be reduced by attaching one or more secondary masses through suitably tuned springs and dampers. The original idea of the so-called tuned mass damper is due to Frahm [1], who proposed supporting the secondary mass by a suitably tuned spring, but without a damper. It was demonstrated by Ormondroyd and Den Hartog [2] that the introduction of a damper not only dissipated energy, but also increased the frequency interval over which the device is effective. This led to development of a design procedure that is still in use. This procedure is based on a single spring supported structural mass equipped with a spring and damper supported secondary mass. The stiffness of the spring supporting the secondary mass is obtained by frequency tuning, based on the observation that there are two frequencies, where the dynamic amplification is independent of the applied damping. Optimal frequency tuning is determined by setting the dynamic amplification at these two neutral frequencies equal, Den Hartog [3]. The selection of the damping level has traditionally been based on an average of (the square of) the vibration absorber damping ratio that would give a local maximum of the dynamic amplification at each of the two neutral frequencies, Brock [4]. The use of an average implies that the resulting damping level does in fact not have the maximum properties used in its derivation. Calibration of the absorber properties based on the actual maximum of the dynamic amplification has been treated by Nihihara et al. [5,6], leading to slightly modified design expressions for both absorber stiffness and absorber damping. Ozer and Royston [7,8] have considered an

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extension of the classic Den Hartog procedure to multi-degree-of-freedom structures by using the Sherman–Morrison formula to reduce the system size, followed by numerical calibration of the individual absorber properties by the classic design procedure [3,4].

A frequency analysis of the tuned mass absorber system by Krenk [9] has demonstrated that the classic frequency tuning leads to equal damping ratio of the two vibration modes of the combined system. Design by equal modal damping has also been used for a tuned mass absorber on a structure exposed to stochastic loading in [10] and for a group of resonant vibration control formats in [11]. It should be noted that the equal modal damping property is in fact lost when the classic frequency calibration is changed, as in [5,6], where the absorber design is based on minimizing the maximum amplification. Furthermore it was demonstrated in [9] that modal damping of the combined system is very closely approximated by half the damping ratio of the vibration absorber. Thus, it is desirable to introduce as much damping as compatible with the frequency response of the combined system. It was found that, at the frequency corresponding to locking the two masses, the relative motion of the secondary mass is independent of the damping, and the optimal damping was defined by equal dynamic amplification at this frequency and at the neutral frequencies. This introduces about 15 percent more damping into the system than the classic value of Den Hartog and Brock [3,4].

A summary of some of the most common design strategies for the classic tuned mass absorber is given in [12], which also proposes a design based on the compromise between reducing the motion of the structure and maximizing the power dissipation in the absorber. The design of the absorber with respect to transient response has been considered in [13], where stiffness and damping are calibrated by a pole-placement technique that maximizes the smallest imaginary part of the complex natural frequencies of the modes associated with the vibration form of interest. The calibration procedures for classic tuned mass absorbers have also been applied to absorbers with various modifications, such as an absorber with both translational and rotation degrees-of-freedom [14], or an absorber where the damper element operates on the absolute, instead of relative, motion of the absorber mass in a sky- or ground-hook format [15].

Most calibration procedures use an idealized structure, characterized by its mass and natural frequency, but without structural damping. Several attempts have been made to extend the theory to more general systems including the structural damping, e.g. by reformulating the frequency response function [16]. Extensive surveys of results in which the effect of initial structural damping is included by series expansions have been presented in [6,17–19]. As seen in for example [20,21] a common feature of these results for the parameters of the tuned mass damper is that they are quite elaborate and for realistic values of the structural damping the effect of the expansion is in fact smaller than the differences following from adopting different optimality criteria, as mentioned above. The ability to obtain analytical results depends critically on the properties of the characteristic equation of the combined structure–absorber system, and it was demonstrated in [10], how approximate but quite accurate explicit results including structural damping could be obtained by small modifications of the coefficients of the characteristic equation.

In the present paper a simple explicit design procedure is developed for a linear tuned mass absorber with a viscous damper on a flexible structure. First, the design procedure is developed based on an idealized two-degree-of-freedom system consisting of a linear vibration absorber mounted on a concentrated spring supported structural mass. The procedure uses the classic frequency tuning. The magnitude of the damping in the absorber is determined from the resulting dynamic amplification of the combined system. This procedure is then extended to cover a tuned mass absorber on a flexible structure. The flexibility of the structure implies that motion of the point at which the absorber is mounted contains a non-resonant contribution in addition to the resonant mode that is to be damped, the so-called additional background flexibility. The background flexibility from non-resonant modes is determined by including a quasi-static correction to a single-mode analysis, a special case of the truncated modal analysis with quasi-static correction developed in [22,23]. It is demonstrated how the background flexibility is identified in terms of a single parameter, and how the effect of background flexibility can be incorporated into the original design procedure for the idealized structure–absorber system by use of equivalent stiffness and damper parameters to represent the absorber. The present method of equivalent absorber parameters is considerably simpler than that presented in [24] for resonant vibration control, using an approximate modified form of the frequency response function. An explicit design procedure based on equivalent absorber parameters is developed, in which the desired level of dynamic amplification is specified. Three numerical examples demonstrate that this design procedure for the tuned mass absorber leads to balanced frequency response curves with amplification levels that closely match the prescribed design value, even for indirect location of the absorber with large influence of the background flexibility.

2. Basic equations of the tuned mass absorber

The basic concept of a tuned mass absorber is illustrated in Fig. 1 showing a structural mass m_s suspended by an elastic spring k_s representing the structural stiffness, and the tuned mass absorber consisting of the mass m_a supported by a spring k_a and a viscous damper c_a acting in parallel. In the design problem the structural parameters m_s and k_s are considered given, and the objective is to determine an optimal set of absorber parameters m_a, k_a and c_a that will provide the desired amplification factor for a harmonic force acting around the resonance frequency.

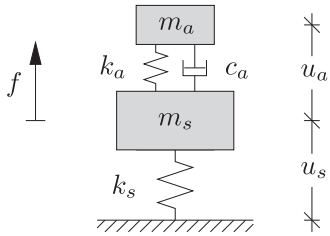

Fig. 1. Tuned mass absorber.

The equations of motion are conveniently derived from the energy balance equation

$$\frac{\mathrm{d}}{\mathrm{d}t}(T+V)=W-D, \tag{1}$$

in which T and V are the kinetic and potential energy, respectively, while W is the external rate of work and D is the rate of dissipation provided by the damper. In the present context it is convenient to describe the motion of the combined system by the motion of the structural mass u_s and the *relative* motion of the absorber mass u_a. These variables give simple equations, and they are also the two important design variables: reduced motion of the structure $u_s(t)$ for comfort and use, and limited relative motion of the absorber mass $u_a(t)$ due to design constraints on the damper (limited space, length of springs, etc.). With these variables to describe the motion the energy terms take the form,

$$\begin{aligned} T&=\tfrac{1}{2}m_s\dot{u}_s^2+\tfrac{1}{2}m_a(\dot{u}_s+\dot{u}_a)^2,\\ V&=\tfrac{1}{2}k_su_s^2+\tfrac{1}{2}k_au_a^2,\\ D&=c_a\dot{u}_a^2. \end{aligned} \tag{2}$$

The equations of motion follow from the energy balance equation:

$$\begin{aligned} m_s\ddot{u}_s+m_a(\ddot{u}_s+\ddot{u}_a)+k_au_s&=f(t),\\ m_a(\ddot{u}_s+\ddot{u}_a)+c_a\dot{u}_a+k_au_a&=0. \end{aligned} \tag{3}$$

The design problem of selecting the absorber parameters m_a, k_a and c_a for given structural mass m_s and stiffness k_s will be considered in the frequency domain.

2.1. Frequency response

The load and response are assumed harmonic with angular frequency ω, corresponding to the representation

$$u_s(t)=u_s\mathrm{e}^{\mathrm{i}\omega t},\quad u_a(t)=u_a\mathrm{e}^{\mathrm{i}\omega t},\quad f(t)=f\mathrm{e}^{\mathrm{i}\omega t}. \tag{4}$$

In the following u_s, u_a and f represent the amplitudes of the displacements and force, respectively. Substitution of the representation (4) into the equations of motion gives

$$\begin{aligned} [k_s-\omega^2(m_s+m_a)]u_s-\omega^2m_au_a&=f,\\ -\omega^2mu_s+[k_a+\mathrm{i}\omega c_a-\omega^2m_a]u_a&=0. \end{aligned} \tag{5}$$

The structure is characterized by its mass m_s and its natural frequency ω_s, while the vibration absorber is characterized by its mass m_a, natural frequency ω_a and damping ratio ζ_a. It is therefore convenient to introduce the parameters

$$\begin{aligned} \omega_s^2&=\frac{k_s}{m_s},\quad \mu=\frac{m_a}{m_s},\\ \omega_a^2&=\frac{k_a}{m_a},\quad \zeta_a=\frac{c_a}{2\sqrt{k_am_a}}, \end{aligned} \tag{6}$$

where the mass ratio μ is a measure of the 'size' of the absorber. These equations are now solved for the amplification factor u_sk_s/f for the structural motion and u_ak_s/f for the relative motion of the absorber mass. Solution of these equations gives the complex displacement amplitudes

$$\frac{u_s}{f/k_s}=\frac{\omega_s^2[\omega_a^2-\omega^2+2\mathrm{i}\zeta_a\omega_a\omega]}{\omega^4-[\omega_s^2+(1+\mu)\omega_a^2]\omega^2+\omega_s^2\omega_a^2+2\mathrm{i}\zeta_a\omega_a\omega[\omega_s^2-(1+\mu)\omega^2]} \tag{7}$$

and

$$\frac{u_a}{f/k_s}=\frac{\omega_s^2\omega^2}{\omega^4-[\omega_s^2+(1+\mu)\omega_a^2]\omega^2+\omega_s^2\omega_a^2+2\mathrm{i}\zeta_a\omega_a\omega[\omega_s^2-(1+\mu)\omega^2]} \tag{8}$$

where the denominator f/k_s is the equivalent static displacement of the structure. In the derivation of the frequency properties of the system the mass ratio μ is selected, and the damper system is then characterized by the frequency ratio

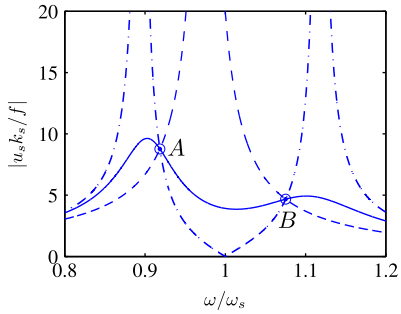

Fig. 2. Dynamic amplification for $\mu=0.05$, $\omega_a=\omega_s$ and $\zeta_a=0$ (- · - · -), 0.1 (——) and ∞ (- - -).

ω_a/ω_s and the damping ratio ζ_a. Optimal damper characteristics can then be identified from a frequency analysis of the two amplification factors.

3. Optimal parameters

If damping is too low, undesirable dynamic amplification will occur near the two undamped resonance frequencies, because the damper does not absorb enough energy. On the other hand, if damping is too high, the damper restricts the relative motion, and thereby creates the effect of a single mass m_s+m_a, oscillating with frequency

$$\omega_\infty=\sqrt{\frac{k_s}{m_s+m_a}}=\frac{\omega_s}{\sqrt{1+\mu}}. \tag{9}$$

The transition from two resonance peaks to one with increased damping is illustrated in Fig. 2.

It is a remarkable fact that there are two frequencies ω_A and ω_B around the natural frequency ω_s for which the magnitude of the response is independent of the damping parameter c_a. These points can be used to determine a suitable tuning frequency ω_a of the damper. The idea is to select the damper frequency ω_a such that the dynamic amplification is equal at the two frequencies ω_A and ω_B [2,3]. The optimal damping ζ_a can then be determined subsequently, and the resulting amplification factors for the structure response u_s and the relative motion of the absorber mass u_a can be calculated. The derivation is summarized in the following.

3.1. Frequency tuning

The dynamic amplification factors at ω_A and ω_B are independent of damping, and can be determined by studying the structure of the frequency response formula (7). Both numerator and denominator consist of a real part that is independent of the damping ratio ζ_a and an imaginary part that has the factor $2\zeta_a$. This is expressed in the form

$$\frac{u_s}{f/k_s}=\frac{A+2\mathrm{i}\zeta_a B}{C+2\mathrm{i}\zeta_a D}. \tag{10}$$

The corresponding dynamic amplification is the absolute value, determined from

$$\left|\frac{u_s}{f/k_s}\right|^2=\frac{A^2+(2\zeta_a)^2B^2}{C^2+(2\zeta_a)^2D^2}. \tag{11}$$

For the magnitude to be independent of ζ_a the limits for $\zeta_a=0$ and $\zeta_a\to\infty$ must be equal, corresponding to $A^2/C^2=B^2/D^2$ or $AD=\pm BC$. Substitution of the terms A, B, C and D gives

$$\left[1-\left(\frac{\omega}{\omega_s}\right)^2\right]\left[1-\left(\frac{\omega}{\omega_a}\right)^2\right]-\mu\left(\frac{\omega}{\omega_s}\right)^2=\pm\left[1-\left(\frac{\omega}{\omega_a}\right)^2\right]\left[1-(1+\mu)\left(\frac{\omega}{\omega_s}\right)^2\right] \tag{12}$$

after cancelling the common factor ω/ω_a. Use of the plus sign leads to the root $\omega=0$. This is the static solution where there is no motion, and therefore no damping force. Use of the minus sign leads to the following quadratic equation in ω^2:

$$(2+\mu)\left(\frac{\omega}{\omega_s}\right)^2\left(\frac{\omega}{\omega_a}\right)^2-2\left[\left(\frac{\omega}{\omega_a}\right)^2+(1+\mu)\left(\frac{\omega}{\omega_s}\right)^2\right]+2=0. \tag{13}$$

The roots of this quadratic equation are ω_A^2 and ω_B^2. The sum of the roots of a quadratic equation is equal to minus the coefficient of the linear term, divided by the coefficient of the quadratic term. In this case

$$\left(\frac{\omega_A}{\omega_s}\right)^2+\left(\frac{\omega_B}{\omega_s}\right)^2=\frac{2}{2+\mu}\left[1+(1+\mu)\left(\frac{\omega_a}{\omega_s}\right)^2\right]. \tag{14}$$

This provides one relation for the frequencies ω_A and ω_B.

The other relation is determined by specifying equal magnitude of the dynamic amplification at the frequencies ω_A and ω_B. At these frequencies the response magnitude is independent of ζ_a, and the relevant response can therefore be determined from (7) for $\zeta_a \to \infty$. This gives the simple result

$$\left|\frac{u_s}{f/k_s}\right|_{A,B}=\frac{1}{1-(1+\mu)\left(\frac{\omega_A}{\omega_s}\right)^2}=\frac{-1}{1-(1+\mu)\left(\frac{\omega_B}{\omega_s}\right)^2}, \tag{15}$$

where the minus sign in the last expression is due to ω_B being larger than ω_∞. Multiplication of the last equality by the product of the denominators gives the equation

$$\left(\frac{\omega_A}{\omega_s}\right)^2+\left(\frac{\omega_B}{\omega_s}\right)^2=\frac{2}{1+\mu}. \tag{16}$$

This is the condition that the dynamic amplification is identical at ω_A and ω_B.

The optimal tuning frequency is now determined by elimination of the sum of squares of the frequencies ω_A and ω_B between Eqs. (14) and (16). This determines the optimal absorber frequency as

$$\frac{\omega_a}{\omega_s}=\frac{1}{1+\mu}. \tag{17}$$

It is seen that optimal frequency tuning of the absorber system corresponds to $\omega_a < \omega_s$ for the present case of a concentrated structural mass. This relation is changed, when considering a tuned absorber on a flexible structure in Section 4.

3.2. Structural amplification factor

The frequency tuning was based on equal dynamic amplification of the structure response u_s at the two frequencies ω_A and ω_B. These frequencies are determined from (13) for the special absorber frequency ω_a given by (17)

$$(2+\mu)(1+\mu)^2\left(\frac{\omega}{\omega_s}\right)^4-2(1+\mu)(2+\mu)\left(\frac{\omega}{\omega_s}\right)^2+2=0. \tag{18}$$

This is a quadratic equation in $\omega_{A,B}^2$ with the solution

$$(1+\mu)\left(\frac{\omega_{A,B}}{\omega_s}\right)^2=1\pm\sqrt{\frac{\mu}{2+\mu}}. \tag{19}$$

Substitution of these frequencies back into the structural amplification formula (15) gives

$$\left|\frac{u_s}{f/k_s}\right|_{A,B}=\sqrt{\frac{2+\mu}{\mu}}. \tag{20}$$

In practice the mass of the damper system is most often considerably less than that of the original structure, and thus the mass ratio μ is much less than one. It is seen that this limits the reduction of dynamic amplification that can be attained at the frequencies ω_A and ω_B. Thus, a mass ratio of $\mu = 0.02$ which is typical in applications leads to a dynamic amplification of the structural response of 10.

3.3. Modal damping

The frequency tuning (17) has the remarkable and useful property that the two vibration modes, generated by the introduction of the absorber on the structure, have identical modal damping ratio, and that this common modal damping ratio is equal to half the absorber damping ratio ζ_a to a very good approximation [9]. When using the frequency tuning (17), the absorber and structure frequencies ω_a and ω_s can be eliminated in favor of the reference frequency ω_∞ from (9), by using the relations

$$\omega_\infty=\frac{\omega_s}{\sqrt{1+\mu}}=\sqrt{\omega_a\omega_s}. \tag{21}$$

Hereby the characteristic polynomial of the frequency response relations (7) and (8) takes the form

$$\omega^4-(2+\mu)\omega_\infty^2\omega^2+\omega_\infty^2-2\mathrm{i}\zeta_a\sqrt{1+\mu}\omega_\infty\omega[\omega^2-\omega_\infty^2]=0. \tag{22}$$

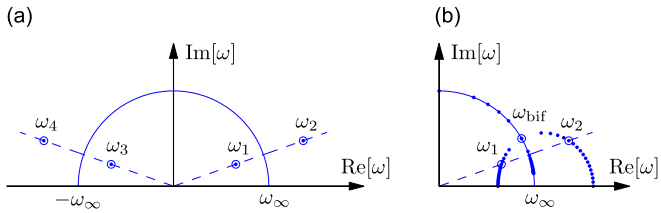

Fig. 3. (a) Complex roots ω_1, ω_2 and ω_3, ω_4 as inverse points of circle $|\omega| = \omega_\infty$. (b) Root-locus diagram with bifurcation point.

After division by $\omega_\infty^2 \omega^2$ this equation can be written in the form of a quadratic equation

$$\left(\frac{\omega}{\omega_\infty} - \frac{\omega_\infty}{\omega}\right)^2 - 2\mathrm{i}\zeta_a\sqrt{1+\mu}\left(\frac{\omega}{\omega_\infty} - \frac{\omega_\infty}{\omega}\right) - \mu = 0. \tag{23}$$

The analytical solution of this equation was derived in [9] by first solving the quadratic equation for the term in the parentheses, and then solving another quadratic equation for ω. The full solution is rather elaborate, and only some basic properties of the solution are needed in the present context.

It is conceptually convenient to consider the problem for the damping parameter ζ_a increasing from zero, thereby generating a root-locus diagram. At zero damping there will be a root $\omega_1 < \omega_\infty$ at the real axis. As damping increases this root moves up into the first quadrant, initially remaining inside the circle with radius ω_∞. It follows from the format of the characteristic equation (23) that if ω_1 is a root, then ω_2 defined by the relation

$$\frac{\omega_2}{\omega_\infty} = \frac{\omega_\infty}{\omega_1^*}, \tag{24}$$

where the asterisk denotes the complex conjugate, is also a root. When ω_1 is inside the circle with radius ω_∞ the root ω_2 is outside and located on the same line through the origin, corresponding to the complex frequencies ω_1 and ω_2 being inverse points with respect to the circle as illustrated in Fig. 3a. This in turn implies that the two modal frequencies have the same modal damping ratio, as expressed by the relation

$$\omega_{1,2} = |\omega_{1,2}|\left(\sqrt{1-\zeta^2} + \mathrm{i}\zeta\right). \tag{25}$$

This 'equal modal damping' property could have been used to determine the absorber frequency ω_a as an alternative to the classic frequency response approach used here. In fact, the equal modal damping can be used as a calibration principle for a range of resonant damping devices as discussed in [25].

For damping levels leaving $\omega_1 \le \omega_\infty$ there is a simple, approximate but quite accurate, relation between the modal damping ratio ζ and the absorber damping ratio ζ_a. It follows from the fact that the coefficient of the cubic term in the characteristic equation (22) is equal to the sum of the four roots shown in Fig. 3. The symmetry of the roots about the imaginary axis implies that

$$\mathrm{Im}\left[\frac{\omega_1}{\omega_\infty} + \frac{\omega_2}{\omega_\infty}\right] = \zeta_a\sqrt{1+\mu}. \tag{26}$$

When using the representation (25) of the modal frequencies in terms of the damping ratio, and the inverse point property, the relation can be expressed as

$$\zeta = \frac{\sqrt{1+\mu}}{\left|\frac{\omega_1}{\omega_\infty}\right| + \left|\frac{\omega_\infty}{\omega_1}\right|}\zeta_a \simeq \frac{1}{2}\zeta_a, \tag{27}$$

where the last approximation is quite good for realistic mass ratios of typically less than 0.05. This damping ratio relation permits a very simple and straightforward design procedure based on desired magnitude of the resulting modal damping, developed in Section 5.

3.4. Damper tuning

The frequency tuning determines the dynamic amplification at the two frequencies ω_A and ω_B, while the absorber damping ratio ζ_a determines the level of the dynamic amplification at other frequencies. The damping ratio ζ_a is typically chosen so as to generate a more or less level plateau for the amplification curve between the frequencies ω_A and ω_B. In the classic approach due to Brock [4] the value of ζ_a^2 was obtained as an arithmetic average of two situations: one with a local maximum of the dynamic amplification at ω_A and another with a local maximum at ω_B. The choice of averaging the square of the damping ratio is pragmatic as it greatly simplifies the result. The result of this procedure has been adopted by Den Hartog [3] and has found widespread use. Here, a slightly larger value of the damping, providing higher modal damping in

the combined system and reduced maximum motion of the absorber mass will be developed, based on a simple direct procedure from [9].

A simple way of obtaining a suitable damping parameter ζ_a consists in observing from the dynamic amplification curves in Fig. 2 that when the amplification at ω_A and ω_B has been adjusted to be equal, a fairly flat plateau can be obtained between these frequencies by selecting a suitable frequency in between and selecting the damping to give the same dynamic amplification at this frequency. A good choice of this intermediate frequency is the frequency ω_∞ defined by (9) corresponding to the two masses joined to one. When using the optimal frequency tuning (17) the frequency ω_∞ is the geometric mean of the structure frequency ω_s and the absorber frequency ω_a as given by (21). At this frequency the structural amplification factor is

$$\frac{u_s}{f/k_s} = \frac{\mu - 2\mathrm{i}\zeta_a\sqrt{1+\mu}}{\mu}. \tag{28}$$

This corresponds to a dynamic amplification determined by

$$\left|\frac{u_s}{f/k_s}\right|^2_{\omega_\infty} = \frac{\mu^2 + (2\zeta_a)^2(1+\mu)}{\mu^2}. \tag{29}$$

By equating the dynamic amplification at ω_∞ to that at ω_A and ω_B, given in (20), the optimal damping ratio is determined as

$$\zeta^2_{\mathrm{opt}} = \frac{1}{2}\,\frac{\mu}{1+\mu}. \tag{30}$$

This is larger than the classic value of Brock [4], used by Den Hartog [3],

$$\zeta^2_{\mathrm{classic}} = \frac{3}{8}\frac{\mu}{1+\mu} \tag{31}$$

The difference in the dynamic amplification of the structural response is illustrated in Fig. 4a. It is seen that the classic value of the absorber damping ratio leads to a dip between the frequencies ω_A and ω_B, where the amplification is the same by design and independent of the damping ratio.

With increasing damping ζ_a the two complex roots ω_1 and ω_2 approach the circle, and eventually meet in a bifurcation point ω_{bif} as illustrated in Fig. 3b. The bifurcation occurs at the value of ζ_a for which the characteristic equation (23), when considered as a quadratic equation in the term in the parentheses, has a double root. This determines the bifurcation value of the damping ratio:

$$\zeta^2_{\mathrm{bif}} = \frac{\mu}{1+\mu}. \tag{32}$$

When the absorber damping ζ_a increases beyond the bifurcation value ζ_{bif}, the roots branch off along the circle with radius ω_∞. The inverse point relation (24) degenerates to a statement regarding a single root on the circle, and while the damping ratio of one root increases, the damping ratio of the other decreases. Thus, the bifurcation damping ζ_{bif} given by (32) provides the highest modal damping. It may be believed that the use of $\zeta_a = \zeta_{\mathrm{bif}}$ would provide an optimal vibration absorber. However, the use of the bifurcation point also implies that the two modes, generated by the introduction of the damper, will have identical frequency, and it turns out that this leads to constructive interference of the response of these two modes. This is illustrated by the dash-dot curve in Fig. 4a showing how the bifurcation value generates a single peak on the structural amplification curve, considerably larger than for the calibration procedures mentioned above.

It can be seen from the formula (8) that the normalized absorber mass amplitude $u_a k_s/f$ is independent of the damping ratio ζ_a at the frequency $\omega = \omega_\infty$ and is equal to

$$\left|\frac{u_a}{f/k_s}\right|_{\omega_\infty} = \frac{1+\mu}{\mu}. \tag{33}$$

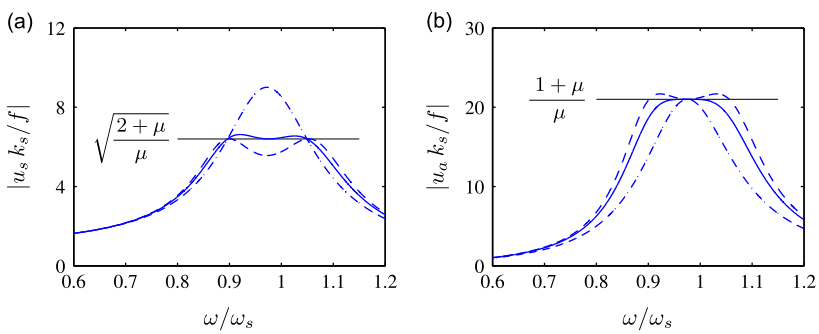

Fig. 4. Dynamic amplification for $\mu = 0.05$: (a) structure motion u_s and (b) relative damper motion u_a. Damping parameter: - -, ζ_{classic}; —— ζ_{opt}; - ·-, ζ_{bif}.

For design reasons it is desirable not to exceed this value. It can be shown by use of (8) that the amplification factor of the relative motion is flat in the sense that its first three derivatives with respect to frequency vanish at this particular frequency, when the damping ratio is ζ_a as given by (30). This is illustrated in Fig. 4b. Any value of ζ_a smaller than ζ_{opt}, such as the classic design value ζ_{classic}, will lead to peaks exceeding the amplification at ω_∞. In contrast, any value of ζ_a larger than ζ_{opt} will lead to a central peak in the dynamic response of the absorber mass.

4. Tuned mass absorber on a flexible structure

Although the basic model of the tuned mass absorber is developed for a concentrated structural mass the main application is its use on flexible structures. Typically the calibration is then based on a selected resonant mode. However, if the structure is sufficiently flexible the tuned absorber will experience support motion not only from the resonant mode, but also from other typically higher frequency modes, and only part of the force delivered by the absorber goes into the targeted resonant mode. This situation is illustrated in Fig. 5 where the total displacement of the tuned absorber support experiences a total displacement u consisting of the resonant part denoted as u_s in the previous sections plus a displacement u_0 from the non-resonant background modes.

Often the non-resonant background modes providing the additional deformation are of higher natural frequency than the targeted resonant mode, and in that case the additional displacement contribution may be considered as being due to an additional flexibility. This is equivalent to the so-called quasi-static correction term often used in connection with dynamic response analysis in terms of a truncated modal expansion, see e.g. Hansteen and Bell [23].

The effect of an additional flexibility between the tuned mass absorber and the structural mass is illustrated in Fig. 6. The effect is represented by connecting the support of the tuned absorber to the structural mass by a spring with stiffness k_0. The identification of the structural mass m_s and the spring stiffness k_s and k_0 from a discrete model of a flexible structure is derived below, followed by the development of a simple method for including the effect of the additional background flexibility $1/k_0$ into a simple modified system, suitable for explicit calibration of the absorber parameters, as discussed in Section 5.

4.1. Background flexibility from non-resonant modes

Tuned mass absorbers are most commonly applied on deformable structures, represented by a finite element model with stiffness matrix $\mathbf{K}$, mass matrix $\mathbf{M}$, displacement vector $\mathbf{u}(t)$, and load vector $\mathbf{f}(t)$. In the context of calibration the structure is represented without structural damping, and the equation of motion of the structure then is

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{f}(t). \tag{34}$$

In the frequency domain this equation takes the form

$$[\mathbf{K}-\omega^2\mathbf{M}]\mathbf{u}=\mathbf{f}, \tag{35}$$

with the solution

$$\mathbf{u}=[\mathbf{K}-\omega^2\mathbf{M}]^{-1}\mathbf{f}. \tag{36}$$

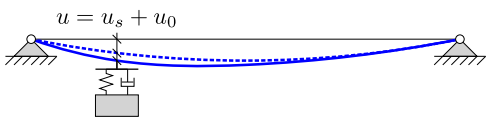

Fig. 5. Resonant and non-resonant displacement components at local vibration absorber.

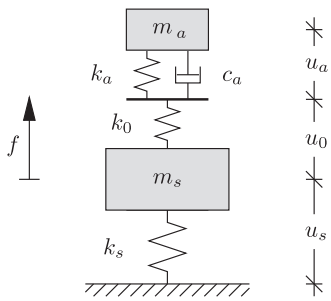

Fig. 6. Modal analogue of tuned mass absorber with flexible support.

In the present context the task is to determine the displacement component u at the location where the mass absorber is mounted to the structure, when a force of magnitude f is acting at this point. This is expressed in terms of the index vector

$$\mathbf{w} = [0, \ldots, 1, \ldots 0]^T, \tag{37}$$

with the unit element placed at the degree of freedom corresponding to the connection with the absorber. In terms of this index vector the displacement and force components are expressed as

$$u = \mathbf{w}^T \mathbf{u} \tag{38a}$$

$$\mathbf{f} = \mathbf{w} f. \tag{38b}$$

Pre-multiplication of (36) by $\mathbf{w}^T$ and substitution of the force representation (38b) then give the scalar response relation

$$u = \mathbf{w}^T [\mathbf{K} - \omega^2 \mathbf{M}]^{-1} \mathbf{w} f. \tag{39}$$

The task now is to obtain an approximate representation of this response relation corresponding to the structure model in Fig. 6 consisting of the structural mass m_s and the two springs with stiffness k_s and k_0.

The approximate representation of the dynamic response relation (39) is obtained by expanding the inverse of the system matrix $[\mathbf{K} - \omega^2 \mathbf{M}]$ in terms of the eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ and the corresponding eigenfrequencies $\omega_1, \ldots, \omega_n$ of the generalized eigenvalue problem corresponding to the frequency equation (35):

$$[\mathbf{K} - \omega^2 \mathbf{M}]^{-1} = \sum_{j=1}^{n} \frac{\omega_j^2}{\omega_j^2 - \omega^2} \frac{\mathbf{u}_j \mathbf{u}_j^T}{\mathbf{u}_j^T \mathbf{K} \mathbf{u}_j}. \tag{40}$$

This result is well known and follows from introducing the modal representation of a general load $\mathbf{f}$ into the modal solution of the frequency equation (35). In the present context of calibration around a frequency ω_s the non-resonant terms are approximated by assuming the corresponding natural frequencies to be larger than ω_s, leading to the approximate expansion

$$[\mathbf{K} - \omega^2 \mathbf{M}]^{-1} \simeq \frac{\omega_s^2}{\omega_s^2 - \omega^2} \frac{\mathbf{u}_s \mathbf{u}_s^T}{\mathbf{u}_s^T \mathbf{K} \mathbf{u}_s} + \sum_{j \neq s} \frac{\mathbf{u}_j \mathbf{u}_j^T}{\mathbf{u}_j^T \mathbf{K} \mathbf{u}_j}, \tag{41}$$

in which the frequency dependence is only retained in the resonant term with frequency ω_s, see e.g. [23].

The summation, and thereby the explicit dependence on the non-resonant modes and frequencies, is eliminated by use of the expansion of the inverse stiffness matrix $\mathbf{K}^{-1}$, following from (40) for $\omega = 0$:

$$\mathbf{K}^{-1} = \sum_{j=1}^{n} \frac{\mathbf{u}_j \mathbf{u}_j^T}{\mathbf{u}_j^T \mathbf{K} \mathbf{u}_j}. \tag{42}$$

When this expansion is used to eliminate the summation in the last expression in (41) the following approximate representation of the inverse of the system matrix is obtained:

$$[\mathbf{K} - \omega^2 \mathbf{M}]^{-1} \simeq \frac{\omega_s^2}{\omega_s^2 - \omega^2} \frac{\mathbf{u}_s \mathbf{u}_s^T}{\mathbf{u}_s^T \mathbf{K} \mathbf{u}_s} + \left(\mathbf{K}^{-1} - \frac{\mathbf{u}_s \mathbf{u}_s^T}{\mathbf{u}_s^T \mathbf{K} \mathbf{u}_s} \right), \tag{43}$$

This approximation is now substituted into the scalar response equation (39), which then gives the total displacement of the structure as

$$u = \left[\frac{\omega_s^2}{\omega_s^2 - \omega^2} \frac{1}{k_s} + \frac{1}{k_0} \right] f. \tag{44}$$

This formula identifies the dynamic structural stiffness k_s as

$$k_s = \frac{\mathbf{u}_s^T \mathbf{K} \mathbf{u}_s}{(\mathbf{w}^T \mathbf{u}_s)^2}, \tag{45}$$

corresponding to the modal stiffness for the normalized mode shape vector $\mathbf{u}_s/(\mathbf{w}^T \mathbf{u}_s)$ with unit modal displacement at the absorber. The corresponding modal mass is

$$m_s = \frac{\mathbf{u}_s^T \mathbf{M} \mathbf{u}_s}{(\mathbf{w}^T \mathbf{u}_s)^2}. \tag{46}$$

The background flexibility $1/k_0$ from the non-resonant modes follows from pre- and post-multiplication of the last term in (43) with $\mathbf{w}^T$ and $\mathbf{w}$, respectively, as

$$\frac{1}{k_0} = \mathbf{w}^T \mathbf{K}^{-1} \mathbf{w} - \frac{(\mathbf{w}^T \mathbf{u}_s)^2}{\mathbf{u}_s^T \mathbf{K} \mathbf{u}_s} = \mathbf{w}^T \mathbf{K}^{-1} \mathbf{w} - \frac{1}{k_s}. \tag{47}$$

It is noted that this relation can be written in the alternative form

$$\frac{1}{k_s} + \frac{1}{k_0} = \mathbf{w}^T \mathbf{K}^{-1} \mathbf{w}. \tag{48}$$

In this form the relation illustrates that the total quasi-static flexibility of the structure, when loaded by a local force at the point of the absorber, consists of the sum of the flexibility $1/k_s$ from the resonant mode and the background flexibility $1/k_0$ from the remaining non-resonant modes.

This completes the determination of the mass m_s and the two stiffness parameters k_s and k_0 associated with the extended single mass structure model shown in Fig. 6. It is to be noted that these parameters are extracted from the stiffness matrix $\mathbf{K}$ and mass matrix $\mathbf{M}$ of the structure using only these matrices and the mode-shape vector $\mathbf{u}_s$ of the resonant mode.

4.2. Effect of background flexibility

A direct frequency analysis of the model representation shown in Fig. 6 will lead to a fifth degree polynomial in the denominator of the frequency response functions for the structural displacement u_s and the absorber displacement u_a, and the fairly simple properties of the original amplification functions would be lost. However, the discrete spring–mass–damper model is only used for calibration of the tuned mass absorber parameters m_a, k_a and c_a, and thus need only be representative in a limited frequency range around resonance. This permits a simple reinterpretation of the original absorber parameters as demonstrated here.

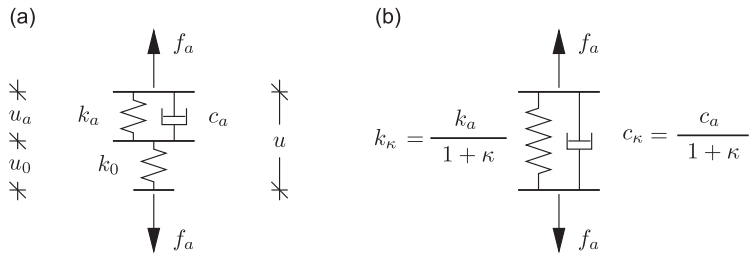

Fig. 7. (a) Spring–damper connection with background flexibility. (b) Connection with equivalent parameters k_κ and c_κ.

The combination of springs and masses representing the full connection between the absorber mass m_a and the structural mass m_s is shown in Fig. 7a. The force transmitted between the two masses is denoted as f_a and the total relative displacement u between the masses is the sum of the background displacement u_0 and the relative displacement of the absorber u_a,

$$u = u_0 + u_a. \tag{49}$$

The absorber force f_a can be expressed both by the background spring k_0 and by the combination of absorber stiffness k_a and viscous element c_a. This gives the relation

$$f_a = k_0 u_0 = k_a u_a + c_a \dot{u}_a. \tag{50}$$

In this equation u_a is eliminated by use of (49), whereby

$$f_a = k_a(u - u_0) + c_a(\dot{u} - \dot{u}_0). \tag{51}$$

Finally, the displacement u_0 is expressed in terms of the force f_a by the first part of (50), whereby

$$\left(1 + \frac{k_a}{k_0}\right) f_a + \frac{c_a}{k_0} \dot{f}_a = k_a u + c_a \dot{u}. \tag{52}$$

Now consider the second term on the left of this relation in the case of harmonic load with angular frequency ω,

$$\frac{c_a}{k_0} \dot{f}_a = \mathrm{i}\omega \frac{c_a}{k_0} f_a = 2\mathrm{i}\zeta_a \frac{k_a}{k_0} \frac{\omega}{\omega_a} f_a. \tag{53}$$

The damping ratio ζ_a is typically not larger than 0.05, and in the calibration frequency range $\omega \simeq \omega_a$. Thus, the second term on the left of (52) will be small compared to the first term, and will be omitted in the following calibration procedure.

When introducing the non-dimensional background flexibility coefficient as

$$\kappa = \frac{k_a}{k_0}, \tag{54}$$

the absorber force transmitted to the absorber mass is then approximated as

$$f_a \simeq k_\kappa u + c_\kappa \dot{u}, \tag{55}$$

with the background flexibility being represented via the equivalent stiffness and viscosity parameters

$$k_\kappa = \frac{k_a}{1+\kappa}, \tag{56a}$$

$$c_\kappa = \frac{c_a}{1+\kappa}. \tag{56b}$$

This approximation, valid in the resonant frequency range and below, corresponds to a connection between absorber mass and structural mass in terms of a parallel coupling of a spring with reduced stiffness k_κ and a damper with reduced viscosity c_κ, as illustrated in Fig. 7b.

5. Design procedure

An effective and simple design procedure for tuned mass absorbers is obtained by using the basic formulation for the classic tuned mass absorber configuration shown in Fig. 1, with the support flexibility included via the equivalent stiffness and damping parameters given in terms of the flexibility coefficient in (56). In the general case the procedure first determines the parameters of the equivalent absorber, and the stiffness and damping parameters are then modified to give the actual absorber parameters. Parameter calibration of a classic tuned mass absorber without support flexibility appears as the special case of $\kappa = 0$.

The level of damping introduced by the tuned mass absorber is conveniently characterized by the dynamic amplification factor around the resonance frequency. In a frequency interval around the structural resonance frequency the dynamic amplification is nearly constant as illustrated in Fig. 4a. The magnitude of the amplification factor in this interval is given by (20). This is conveniently rewritten by use of the damper tuning expression (30) as

$$\left|\frac{u_s}{f/k_s}\right|_{A,B} = \sqrt{\frac{1}{\zeta_{\text{opt}}^2} - 1} \simeq \frac{1}{\zeta_{\text{opt}}} \simeq \frac{1}{2\zeta}, \tag{57}$$

where the last equality follows from the modal damping relation (27). This expression for the dynamic amplification factor is the same as for a system with internal damping corresponding to the modal damping ratio ζ. In the case of a system with a tuned absorber parametric studies indicate that the total equivalent damping ratio is well represented by the sum of the internal structural damping ratio ζ_s and the additional contribution $\frac{1}{2}\zeta_a$ from a vibration absorber with optimal frequency tuning, see [25] for a discussion and numerical examples. In the presence of background flexibility the actual absorber ratio is replaced by the equivalent absorber damping ratio ζ_κ, and thus the design amplification can be expressed in terms of the design modal damping ratio

$$\zeta_d = \tfrac{1}{2}\zeta_\kappa + \zeta_s. \tag{58}$$

In this formula ζ_κ is the equivalent damping ratio of an absorber with a flexible connection to the structural mass, included in the equivalent absorber stiffness and damping parameters k_κ and c_κ, given by (56). The design value of this equivalent

Table 1
Design procedure for tuned mass absorber.

(1)	Equivalent damping: $\zeta_\kappa = 2(\zeta_d - \zeta_s)$.
(2)	Mass ratio: $\mu = \dfrac{2\zeta_\kappa^2}{1-2\zeta_\kappa^2}$.
(3)	Equivalent absorber stiffness: $k_\kappa = \dfrac{\mu}{(1+\mu)^2} k_s$.
(4)	Background flexibility factor: $\kappa = \dfrac{k_\kappa}{k_0 - k_\kappa}$.
(5)	Absorber natural frequency: $\omega_a = \sqrt{1+\kappa}\,\dfrac{\omega_s}{1+\mu}$.
(6)	Absorber damping ratio: $\zeta_a = \sqrt{1+\kappa}\sqrt{\dfrac{\mu}{2(1+\mu)}}$.
(7)	Absorber mass, stiffness and damping coefficient: $m_a = \mu m_s,\ k_a = \omega_a^2 m_a,\ c_a = 2\zeta_a\omega_a m_a$.

damping ratio is given by (58) as

$$\zeta_\kappa = 2(\zeta_d - \zeta_s). \tag{59}$$

In the equivalent vibration absorber model in Fig. 7b the support flexibility $1/k_0$ is absorbed into the parallel spring and mass as in the basic structure–absorber model in Fig. 1, and the mass ratio therefore follows from (30) in terms of the equivalent damping ratio ζ_κ as

$$\mu = \frac{2\zeta_\kappa^2}{1-2\zeta_\kappa^2}. \tag{60}$$

The equivalent system retains structure and absorber masses, and the absorber mass m_a then follows directly from the mass ratio as $m_a = \mu m_s$. The design procedure is summarized in Table 1, where calculation of the equivalent damping ratio ζ_κ and the mass ratio μ constitutes the first two steps.

The frequency ω_κ is now determined as the equivalent absorber frequency by the relation (17) for the equivalent absorber

$$\omega_\kappa^2 = \frac{k_\kappa}{m_a} = \left(\frac{\omega_s}{1+\mu}\right)^2. \tag{61}$$

This relation determines the equivalent absorber stiffness

$$k_\kappa = \frac{\mu}{(1+\mu)^2} k_s. \tag{62}$$

The physical absorber stiffness k_a is related to the equivalent absorber stiffness k_κ by (56a), which accounts for the background flexibility. This relation is written in the form

$$k_\kappa = \frac{k_a}{1+\kappa} = \frac{k_a}{1+k_a/k_0}. \tag{63}$$

Solution of this equation for the physical absorber stiffness k_a leads to the following expression for the background flexibility factor

$$\kappa = \frac{k_a}{k_0} = \frac{k_\kappa/k_0}{1-k_\kappa/k_0} = \frac{k_\kappa}{k_0-k_\kappa}. \tag{64}$$

It is noted that the presence of background flexibility limits the capacity of the damper to $k_\kappa < k_0$, a condition that does not appear to impose a limitation in practice.

With the flexibility factor determined by (64) the absorber stiffness follows from (56a) as

$$k_a = (1+\kappa)k_\kappa = (1+\kappa)\frac{\omega_s^2}{(1+\mu)^2} m_a. \tag{65}$$

The vibration absorber frequency ω_a can then be expressed in terms of the structural frequency ω_s as

$$\omega_a = \sqrt{1+\kappa}\,\omega_\kappa = \sqrt{1+\kappa}\frac{\omega_s}{1+\mu}. \tag{66}$$

The effect of the background flexibility on the absorber design frequency is accounted for via κ by the first factor, and it is seen that support flexibility will lead to an increase of the optimal absorber frequency ω_a, which may in fact become larger than the structural frequency ω_s.

The absorber damping is characterized by the damping ratio ζ_a. It follows from the definition of the damping ratio and the definition (56b) of the relative damping that

$$\frac{\zeta_a\omega_a}{\zeta_\kappa\omega_\kappa} = \frac{c_a}{c_\kappa} = 1+\kappa. \tag{67}$$

The frequency ratio follows from the stiffness ratio (56), and when introducing ζ_κ from (60), the absorber damping ratio is

$$\zeta_a = \sqrt{1+\kappa}\,\zeta_\kappa = \sqrt{1+\kappa}\sqrt{\frac{\mu}{2(1+\mu)}}. \tag{68}$$

Again, the background flexibility is accounted for by the first factor.

It is noted that the mass ratio needed to obtain a given level of damping is independent of background flexibility, while both the absorber frequency ω_a and the absorber damping ratio ζ_a are to be increased by the factor $\sqrt{1+\kappa}$.

6. Examples

The calibration procedure for a tuned mass absorber presented in the previous section is here illustrated by three examples: a simple shear fame that illustrates the main features of the problem, a taut cable with large influence from background flexibility, and a three-span bridge where the presence of a moving part and associated machinery leads to

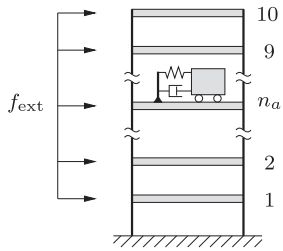

Fig. 8. A 10-story shear frame structure with tuned mass damper on floor n_d.

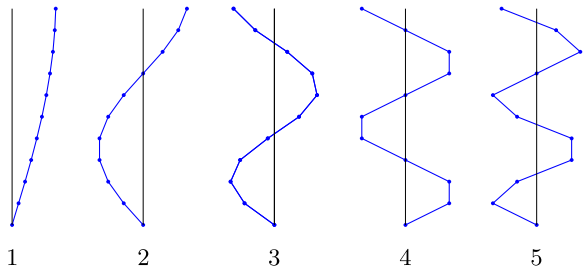

Fig. 9. First five vibration modes of a 10-story shear frame.

Table 2
TMD in shear frame structure. Desired damping $\zeta_d = 0.05$.

n_a	$\frac{m_s}{m_{\text{tot}}}$	$\frac{k_s}{k_0}$	κ	$\frac{m_a}{m_{\text{tot}}}$	$\frac{\omega_a}{\omega_s}$	ζ_a	ζ	$\zeta_{\kappa=0}$
10	0.5279	0.1793	0.0035	0.0108	0.9817	0.1002	0.0503	0.0510
							0.0502	0.0492
8	0.6059	0.0827	0.0016	0.0124	0.9808	0.1001	0.0504	0.0507
							0.0504	0.0499
6	0.8589	0.1512	0.0030	0.0175	0.9815	0.1001	0.0503	0.0509
							0.0503	0.0495
4	1.6544	0.4783	0.0095	0.0338	0.9846	0.1005	0.0500	0.0520
							0.0498	0.0473
2	6.0428	1.6997	0.0345	0.1233	0.9967	0.1017	0.0487	0.0551
							0.0482	0.0404

the tuned mass absorbers being placed more indirectly in the side-spans, thereby introducing the effect of background flexibility.

6.1. Shear frame

The shear frame illustrated in Fig. 8 is a much used example to illustrate calibration and performance of tuned mass absorbers and other damping devices. The figure shows a 10-story shear frame structure, with floor mass m and inter-story stiffness k. The horizontal displacement of the individual floors represent the 10 degrees of freedom.

The tuned mass absorber is located at floor n_a, as shown in Fig. 8, and the example illustrates the influence of the location of the absorber on the optimal device parameters, when the objective is to introduce damping into the first mode. The first five mode shapes are illustrated in Fig. 9. It follows from the first two steps of the design procedure in Table 1 that the design value of the damping ratio ζ_d, determined from the design amplification factor, determines the mass ratio μ. In a multi-degree-of-freedom system like the present the structural mass is the modal mass associated with unit modal displacement at the location of the damper. With this normalization the modal mass, and thereby the necessary absorber mass will increase, as the absorber location is moved down in the building.

In the present example the structural damping in the building is considered negligible, $\zeta_s \simeq 0$, and the objective is to attain a dynamic amplification factor of 10. According to (57) this corresponds to a design value of the resulting modal

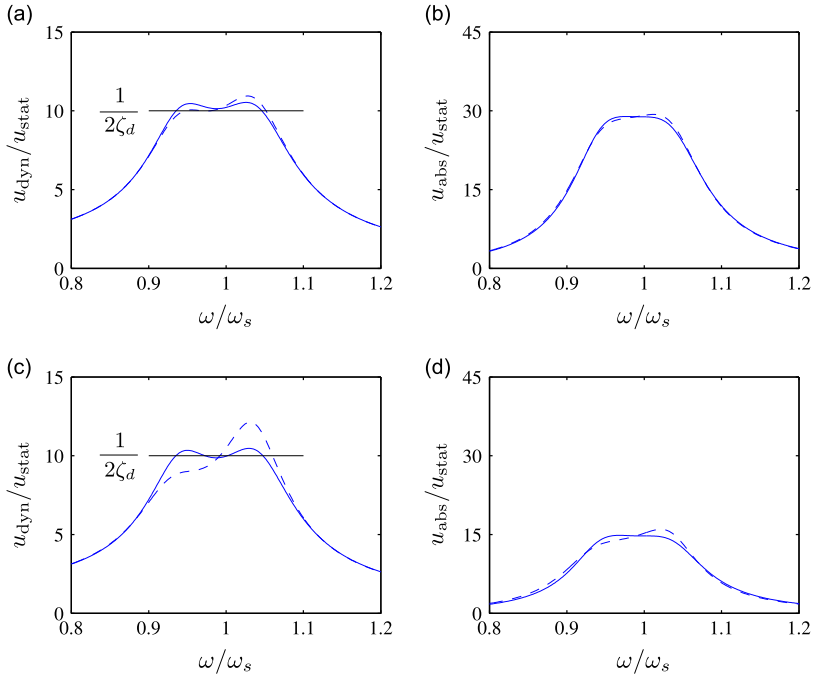

Fig. 10. Amplitude of (a,c) top floor and (b,d) relative damper motion for $\zeta_d = 0.05$. Damper located at n_d=4 (a,b) and 2 (c,d).

damping ratio of $\zeta_d = 0.05$. In the absence of background flexibility, $\kappa = 0$, this would correspond to the absorber damping ratio $\zeta_a = 2\zeta_d = 0.10$, a modal mass ratio of $\mu = 0.0204$, and the absorber frequency ratio $\omega_a/\omega_s = 0.9800$.

The results from a calibration including background flexibility according to the procedure in Table 1 are summarized in Table 2 for a tuned mass absorber located at the top floor, and subsequently two floors lower, etc. as indicated in the first column. The modal mass of the structure m_s is obtained by normalizing the mode shape to unity at the location of the absorber n_a and therefore increases as shown in the second column. Columns three and four show the background flexibility $1/k_0$, relative to the modal structural mass k_s/k_0, and the absorber support flexibility $\kappa = k_a/k_0$, respectively. The background flexibility constitutes the flexibility associated with the non-resonant modes, and it is seen that there is a local minimum around the absorber location n_a=8. This corresponds to a location close to nodes of the modes 2 and 3, and the calibration and performance are nearly independent of these modes for n_a=8. As the absorber is moved down its effect on the first mode becomes more indirect, and although the modal mass ratio μ remains close to 0.02, the normalization of the mode changes, whereby the absorber mass m_a increases as shown in column five.

The final result of the calibration is the modal damping ratio ζ in the two modes in which mode 1 is split by the introduction of the tuned mass absorber. These two damping ratios are shown in the last two columns of Table 2. The first of these columns gives results when including the background flexibility through a finite value of κ, while the last column shows the damping ratios when the influence of background flexibility is omitted, corresponding to $\kappa = 0$. The goal is a balanced design, in which both damping ratios are equal to 0.05. It is seen that this is attained to within a fraction of a percent for the general procedure including the κ-parameter, while the classic design with $\kappa = 0$ becomes increasingly unbalanced as the absorber is moved down, ending with a relative difference between the damping ratio of the two modes of about 30 percent.

The effect of the tuned mass absorber and its calibration is evaluated by performing a direct frequency analysis of the structure including the tuned mass absorber, i.e. without using modal analysis. The load is uniformly distributed as shown in Fig. 8 and of harmonic time variation with angular frequency ω. The top floor is selected as reference and the static displacement u_{stat} is used for normalization. The effect of the tuned mass absorber is illustrated in Fig. 10 showing the dynamic response of the top floor u_{dyn} and the absorber displacement u_{abs} for a tuned mass absorber located at the fourth floor and at the second floor, respectively. It is seen that the background flexibility is of only modest influence for an absorber at the fourth floor, while calibration of an absorber located at the second floor will become severely unbalanced, if neglecting the background flexibility, while including this effect via the flexibility parameter κ leads to a fully balanced design. The horizontal dashed line indicates the theoretical amplification factor $1/2\zeta_d = 10.0$ based on the design value $\zeta_d = 0.05$ of the modal damping ratio. The agreement is seen to be excellent. The motion of the absorber mass is normalized with respect to the top floor displacement u_{stat}, and can therefore be compared directly to the dynamic top floor displacement, while the simple amplification factor indicated in Fig. 4 is based on quasi-static structure response to a local load.

6.2. Taut cable

There is a considerable literature on damping of cables by means of a local damper located close to one end of the cable. In the case of a viscous damper located close to one end of the cable the maximum attainable damping ratio in the cable with damper is determined by the distance a of the damper from the support as a/l, where l is the length of the cable [25]. The use of a viscous damper requires external support, and a typical distance from the cable support is about $a/l \simeq 0.02$, corresponding to a maximum resulting modal damping ratio $\zeta \simeq 0.01$.

The equivalent problem in which the viscous damper is replaced by a tuned mass absorber is illustrated in Fig. 11, and has been treated in e.g. [26–29] without specific consideration of the effect of background flexibility, and in [24] with a somewhat more complicated adjustment for background flexibility based on modification of the frequency response function of the combined system. This configuration is here used to demonstrate the effect of background flexibility on the calibration and performance of the absorber for damping the lowest mode, and the effect of changing the distance of the absorber from the cable support.

Three locations are considered, $a/l = 0.02, 0.04, 0.08$, corresponding to doubling the typical distance $a/l \simeq 0.02$ for a viscous damper. The cable is represented by 50 linear finite elements, whereby the locations $a/l = 0.02, 0.04, 0.08$ correspond to locating the absorber at nodes 1, 2 and 4, respectively. The results of the calibration procedure from Table 1 with and without background flexibility are presented in Table 3. The second column gives the relative modal structural mass for the three locations of the absorber. If the same mode-shape vector were used the numbers would be identical, and the difference is solely due to the normalization with the displacement at the location of the absorber. All three locations are so close to the support that the shape of the resonant mode is approximately linear, and thus doubling the distance from the support will reduce the modal displacement to half. As the modal mass is defined using the square of the modal displacement, doubling the distance of the absorber from the support reduces the corresponding modal mass by a factor of 1/4 as illustrated by the table. Columns 3 and 4 show the background flexibility relative to the structural and the absorber flexibility, respectively. Due to the closeness to the cable support the variation is regular and monotonic, corresponding to a factor of approximately 0.5 when doubling the absorber distance to the cable support.

The absorber mass, given in column five, is determined by the modal mass ratio which is $\mu \simeq 8\zeta_d^2 \simeq 0.0008$ for all three modes, and thus the absorber mass exhibits the same variation as the structural modal mass m_s. This mass ratio is so small that the absorber frequency, determined by (66), becomes larger than the modal frequency of the structure ω_s. This is a clear difference from the classic frequency calibration in [3,9], where the absorber frequency is always lower than the natural frequency of the structure. This frequency adjustment is the key factor in balancing the modal damping ratios of the double modes generated by the tuned absorber and balancing the resonance peaks corresponding to these modes. It is shown in the two last columns of Table 3, how use of the full calibration procedure including the background flexibility balances the damping ratio of the double modes that would otherwise exhibit a relative difference of as much as about 75 percent for the absorber location closest to the cable support. The effect on the dynamic amplification is illustrated in Fig. 12 showing results for a uniformly distributed load with harmonic time variation. It is seen that the unbalance of the two local peaks of the dynamic amplification curve is very dramatic for the absorber location $a/l = 0.02$, and still noticeable for $a/l = 0.08$.

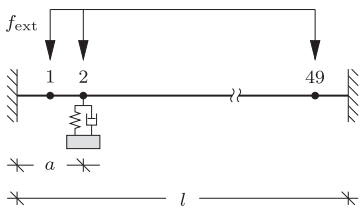

Fig. 11. Taut cable with tuned mass absorber at relative distance a/l.

Table 3
Tuned mass absorber on taut cable. Design damping $\zeta_d = 0.01$.

a/l	$\frac{m_s}{m_{\text{tot}}}$	$\frac{k_s}{k_0}$	κ	$\frac{m_a}{m_{\text{tot}}}$	$\frac{\omega_a}{\omega_s}$	ζ_a	ζ	$\zeta_{\kappa=0}$
0.02	126.82	23.524	0.0192	0.1015	1.0087	0.0202	0.0099	0.0136
							0.0097	0.0059
0.04	31.830	11.059	0.0089	0.0255	1.0036	0.0201	0.0100	0.0120
							0.0098	0.0077
0.08	8.0845	4.8707	0.0039	0.0065	1.0012	0.0200	0.0101	0.0110
							0.0099	0.0089

A similar effect of frequency adjustment was demonstrated in [24], using a somewhat different procedure based on an approximate modification of the frequency response function. The horizontal line in the figure indicates the theoretical amplification level $1/2\zeta_d = 50$.

It should be noticed from Fig. 12 that, while moving the absorber toward the center of the cable increases the efficiency by reducing the absorber mass, the local displacement amplitude goes up from $u_{\text{abs}}/u_{\text{dyn}} \simeq 1.6$ for $a/l = 0.02$ to $u_{\text{abs}}/u_{\text{dyn}} \simeq 6.5$ for $a/l = 0.08$.

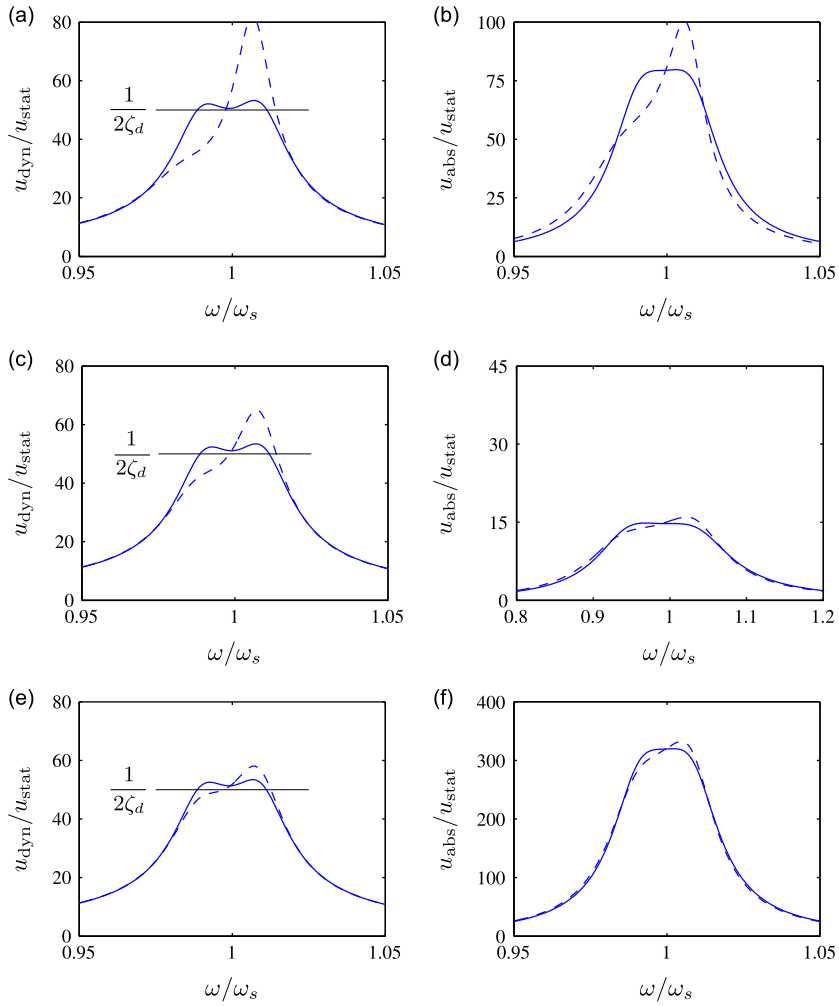

Fig. 12. Amplitude of (a,c,e) cable center and (b,d,f) relative damper motion for $\zeta_d = 0.01$. Damper located at $a/l = 0.02$ (a,b), 0.04 (c,d), and 0.08 (e,f).

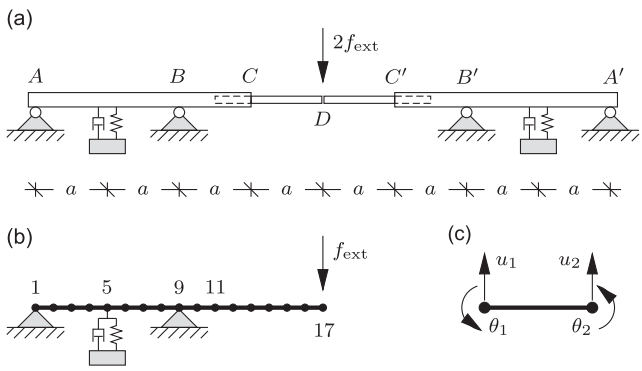

Fig. 13. Symmetric three span pedestrian bridge with movable center span CC'.

6.3. Bridge structure

Fig. 13a shows a three-span footbridge where the central span has the double length of the two side spans. The central part of the bridge consists of two retractable parts, meeting at the center. Due to the mechanism and to leave room for the retractable parts the possibility of placing a set of two symmetrically located tuned mass absorbers in the side spans is investigated. The load is idealized as a force with amplitude $2f_{\text{ext}}$ acting at the center, and due to the symmetry only the left part of the structure is modeled, using 16 beam elements as shown in Fig. 13b,c.

An analysis is performed to compare two levels of damping, $\zeta_d = 0.02$ and $\zeta_d = 0.03$, and two locations of the tuned mass absorber: at node 5 in the middle of the side span, and at node 11 in the cantilevered part of the center span. The results are shown in Table 4. The ratio m_s/m_{tot} of the modal mass and the total mass of the bridge deck shown in column three clearly indicates that the position in the side span is somewhat more indirect. Also, as shown in column four, the background flexibility is considerably larger for the side span location. The design procedure of Table 1 leads to an absorber mass ratio m_a/m_{tot} that is about three times larger for the side span position.

For the side-span location of the tuned mass absorbers the damping ratio of the two modes generated by the split of the lowest mode by the introduction of the absorber exhibit a relative magnitude of 21 percent and 14 percent, respectively, if the effect of background flexibility is neglected. This difference is balanced completely by including the background flexibility, and the design damping ratio ζ_d is attained to within about 1 percent. For the center-span location neglecting the background flexibility only leads to a relative difference of the two modal damping ratios of about 1 percent, while the full design procedure leads to a fully balanced design resulting in the design damping ratio ζ_d being imposed on both modes.

Table 4
TMD on pedestrian bridge.

n_a	ζ_d	$\frac{m_s}{m_{\text{tot}}}$	$\frac{k_s}{k_0}$	κ	$\frac{m_a}{m_{\text{tot}}}$	$\frac{\omega_a}{\omega_s}$	ζ_a	ζ	$\zeta_{\kappa=0}$
5	0.02	13.316	1.8534	0.0059	0.0427	0.9998	0.0401	0.0199	0.0213
								0.0199	0.0184
	0.03			0.0134	0.0966	0.9994	0.0604	0.0297	0.0326
								0.0296	0.0263
11	0.02	4.5001	0.2054	0.0007	0.0144	0.9971	0.0400	0.0200	0.0202
								0.0200	0.0198
	0.03			0.0015	0.0326	0.9935	0.0600	0.0301	0.0304
								0.0300	0.0297

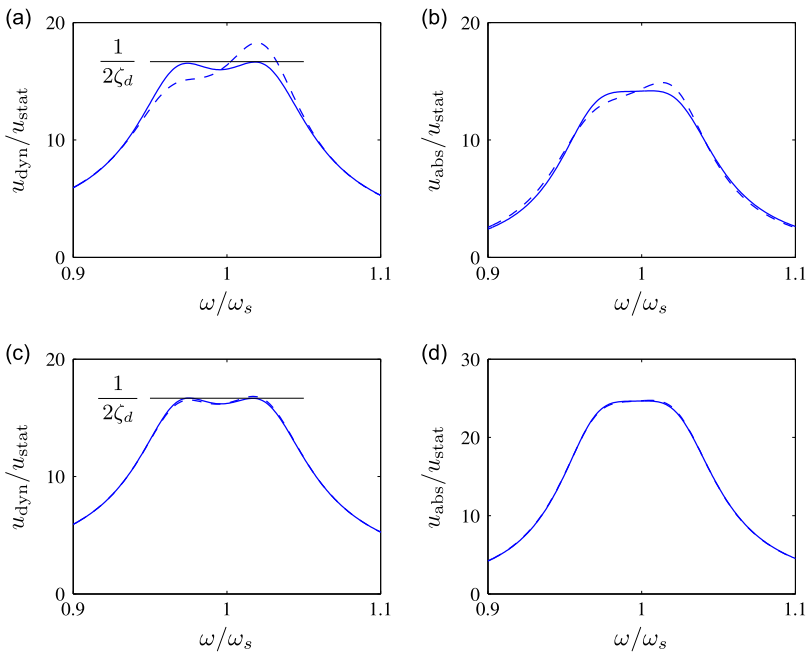

Fig. 14. Amplitude of beam tip motion (a,c) and relative damper motion (b,d) for $\zeta_d = 0.03$. Damper located at node 5 (a,b) and at node 11 (c,d).

It is noticed that for the location of the absorber in the side-span the absorber frequency ratio is $\omega_a/\omega_s \simeq 1$. As demonstrated by the results in the previous examples, and the results for location of the absorber in the center-span, this is purely a coincidence and is not a suitable design criterion. In fact, according to (66) the absorber frequency ratio ω_a/ω_s associated with a balanced design depends on the mass ratio μ as well as the background flexibility parameter κ, and does not have a universal value.

Finally, the dynamic response of the center point D as evaluated from a direct harmonic response analysis for the case $\zeta_d = 0.03$ is shown in Fig. 14. As already indicated by the results for the modal damping ratios the side-span location leads to a noticeable unbalance that is completely corrected by including the background flexibility in the design procedure, while the response with the absorber in the center-span location is fully balanced, irrespective of inclusion of the background flexibility effect. In both cases the actual response plateau is well represented by the design value $1/2\zeta_d = 16.66$, shown by the thin horizontal line in the figure. It is observed that the response amplitude of the absorber is reduced by a factor of about 0.6 when the absorber is moved from the center-span to the side-span location.

7. Conclusions

Tuned mass absorbers are typically used to introduce damping in specific vibration modes of flexible structures with distributed mass. The tuned absorber is mounted locally, and therefore its motion consists partly of the specific modal displacement and partly of motion from other, typically non-resonant, modes. The contribution to the absorber displacement from these other modes typically takes the form of an apparent additional flexibility of the absorber support. Hereby the classic calibration of the tuned mass absorber based on an idealized concentrated structural mass will lead to errors in the optimality of the absorber stiffness and damping parameters. These errors would lead to an unbalance in the frequency response of the combined structure–absorber system with a higher dynamic amplification at the higher of the two natural vibration frequencies, in which the resonant vibration mode is split by the introduction of the tuned absorber. In this paper a simple procedure has been devised in which the stiffness and damping in the tuned mass absorber are represented in a form that incorporates the effect of the background flexibility in an approximate but highly accurate manner via the background flexibility parameter, expressing the ratio of the spring stiffness of the vibration absorber to the structural background stiffness. The mass of the absorber is typically a few percent of the relevant structural mass, and therefore the background flexibility parameter is much less than unity. This renders the method very accurate, in spite of the approximation inherent in the introduction of equivalent absorber stiffness and damping parameters.

An explicit design method has been developed based on the equivalent absorber parameters and the structure represented by the relevant mode shape and the associated modal mass and stiffness, plus an additional quasi-static stiffness associated with the remaining modes. The design procedure starts from the design value of the maximum dynamic amplification factor, and then proceeds with calculation of the absorber parameters by a simple sequence of analytical expressions summarized in Table 1. The background flexibility leads to higher design values for the absorber frequency and damping parameter. The background flexibility plays an important role, when the vibration absorber is located where other modes make considerable contributions to the displacements, such as in cables or when placing the absorber in side-spans of bridges. In all the considered cases the design procedure has led to a fully balanced frequency curve for the dynamic amplification of the structural response and to maximum amplification values corresponding closely to the initial design values.

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